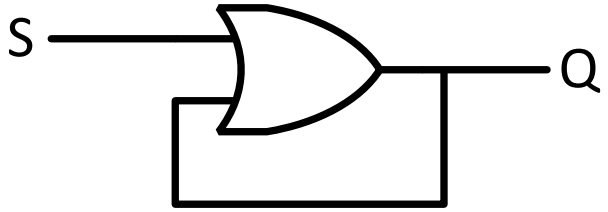
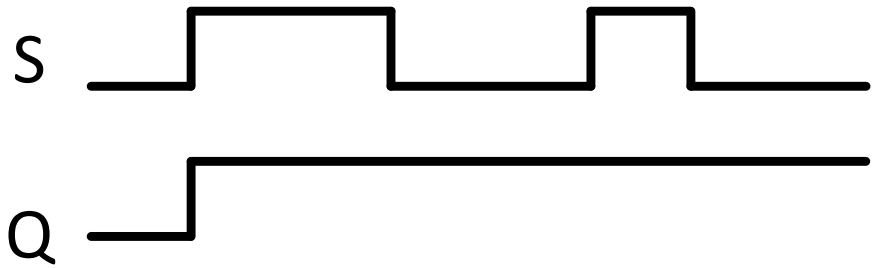
Circuitos Sequenciais – Latch, Flip-Flop

Circuito Sequencial

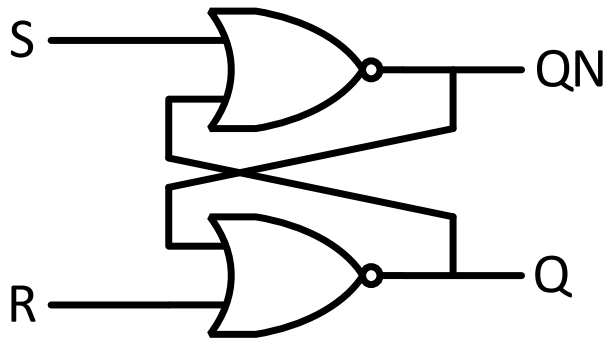
- Circuito digital:
  - Combinatório – Saídas dependem apenas das entradas presentes;
  - **Sequencial** – Saídas dependem não só das entradas presentes, como também da sequência de entradas anterior.



S	Q_t	Q_{t+1}
0	0	0
0	1	1
1	0	1
1	1	1



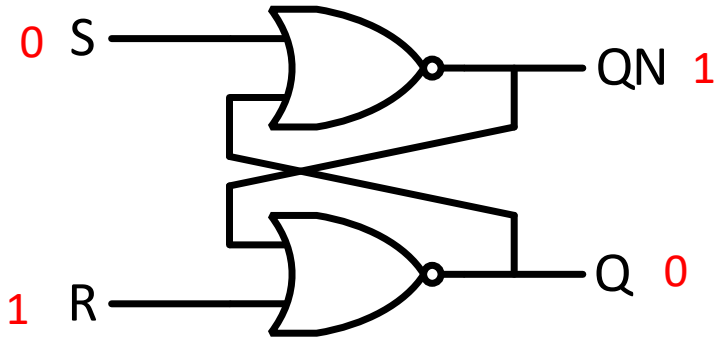
Latch S-R (com portas NOR)

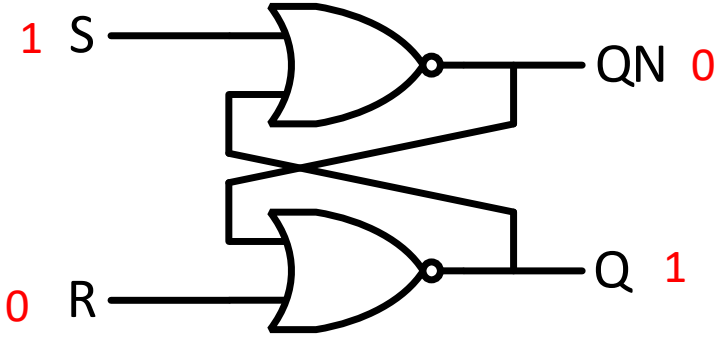


S	R	Q	QN
0	0	Q	QN
0	1	0	1
1	0	1	0
1	1	0	0

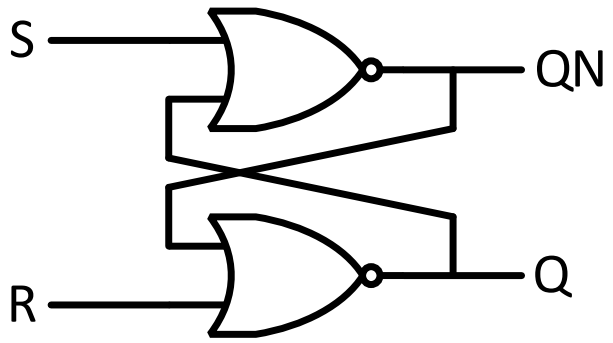
S – Set

R - Reset





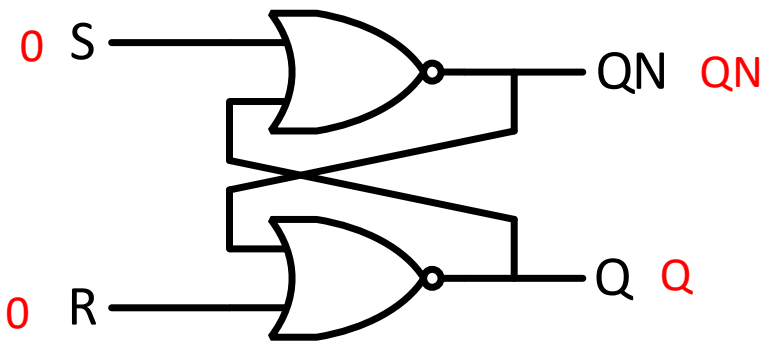
Latch S-R (com portas NOR)

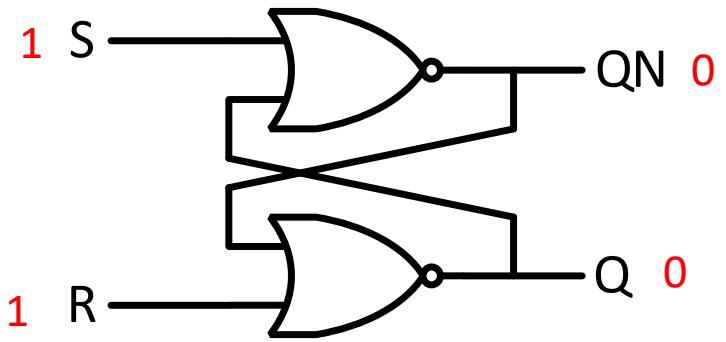


S	R	Q	QN
0	0	Q	QN
0	1	0	1
1	0	1	0
1	1	0	0

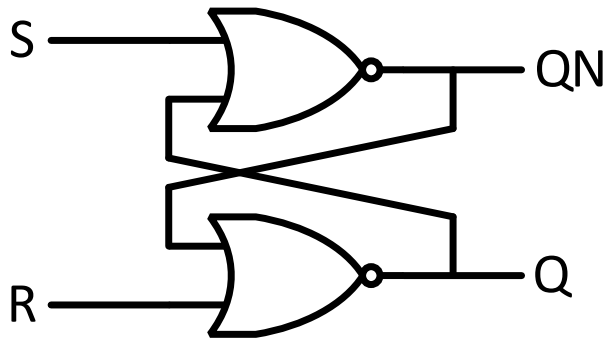
S – Set

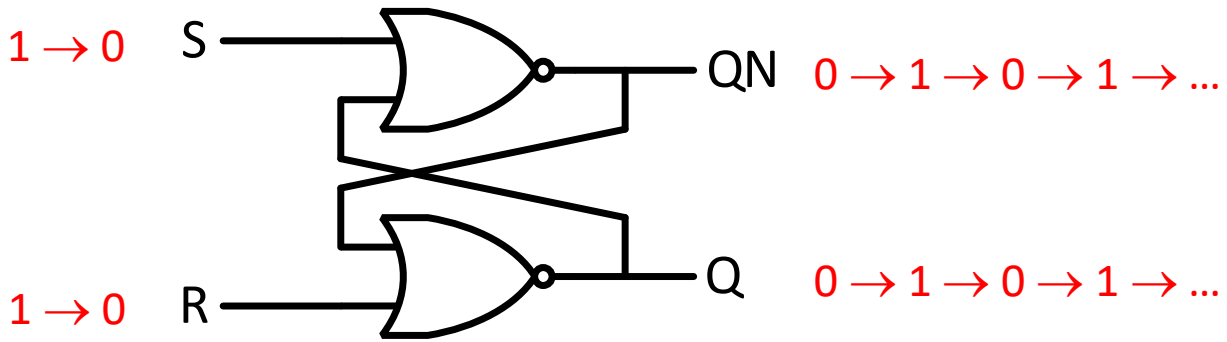
R - Reset





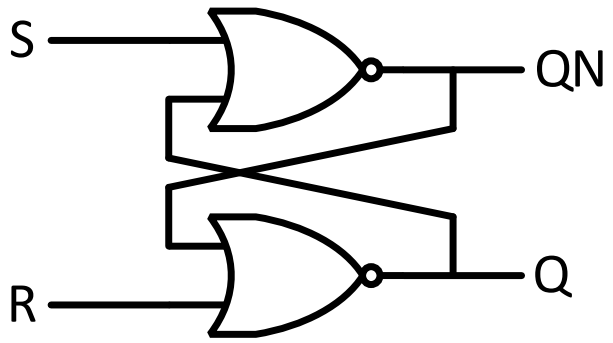
Latch S-R (com portas NOR)



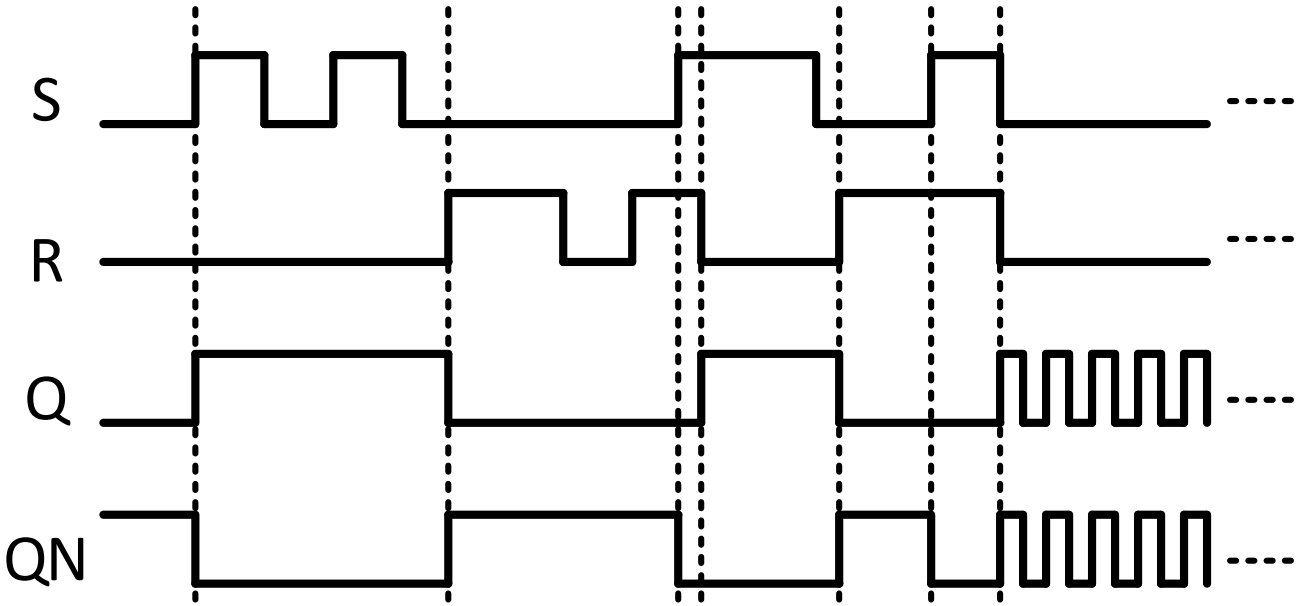


S	R	Q	QN
0	0	Q	QN
0	1	0	1
1	0	1	0
1	1	0	0

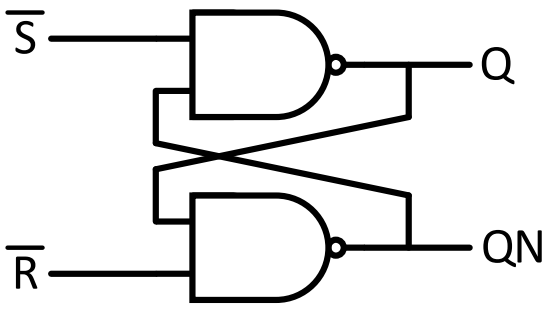
Latch S-R (com portas NOR)



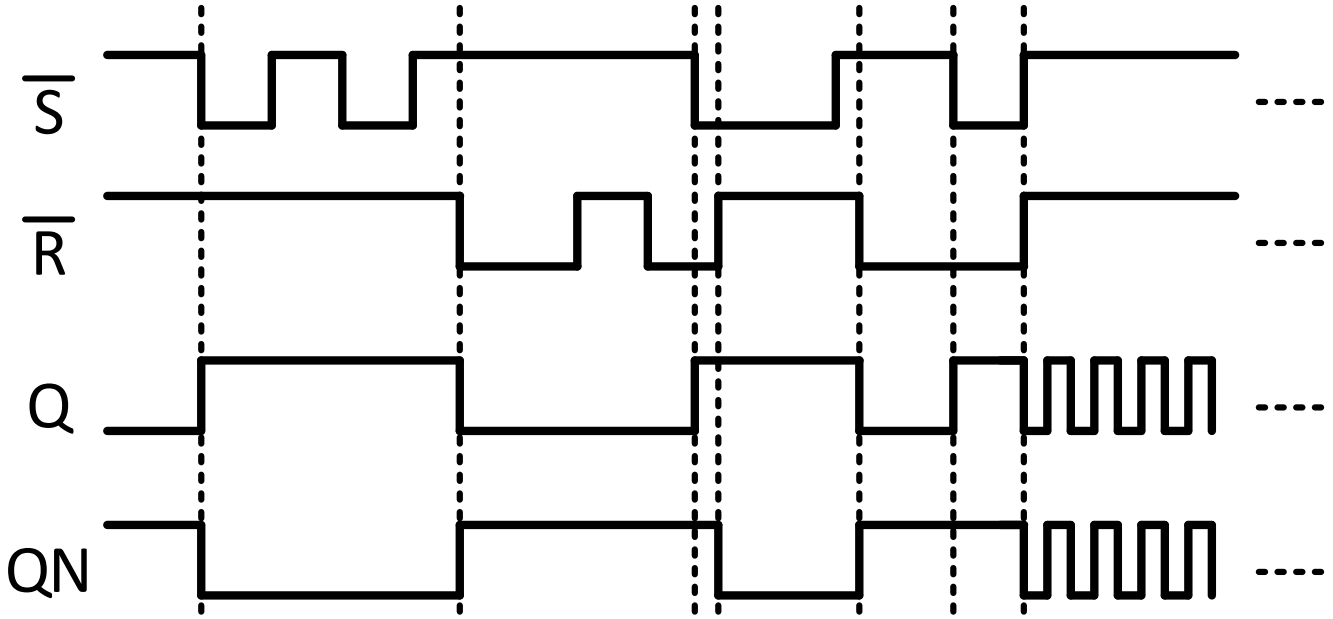
S	R	Q	QN
0	0	Q	QN
0	1	0	1
1	0	1	0
1	1	0	0



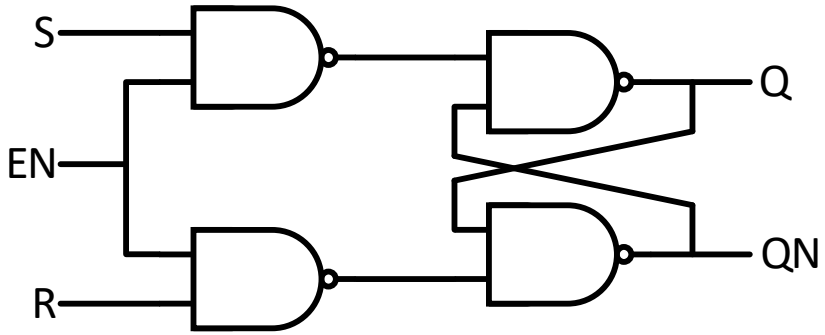
Latch S-R (com portas NAND)



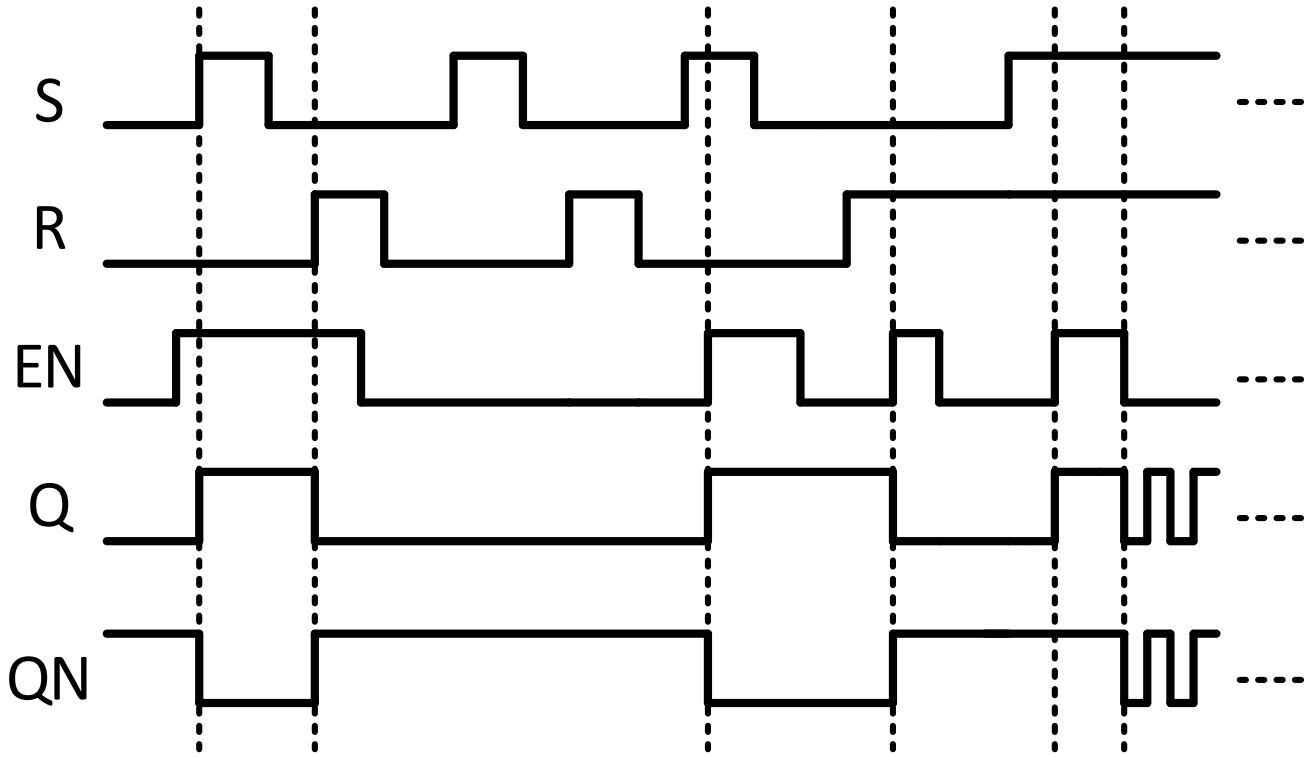
$\overline{S}$	$\overline{R}$	Q	QN
0	0	1	1
0	1	1	0
1	0	0	1
1	1	Q	QN



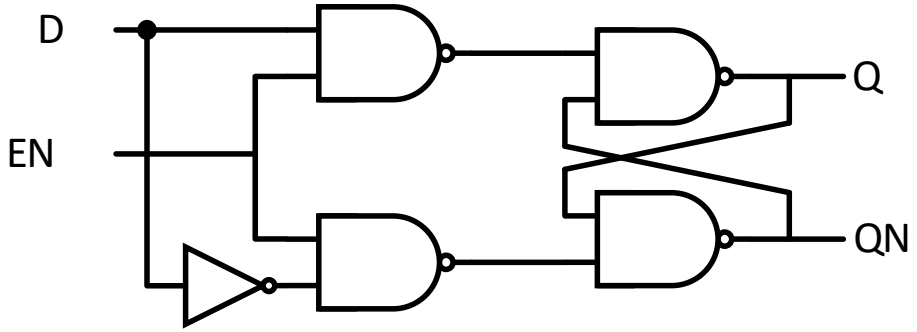
Latch S-R com *Enable*



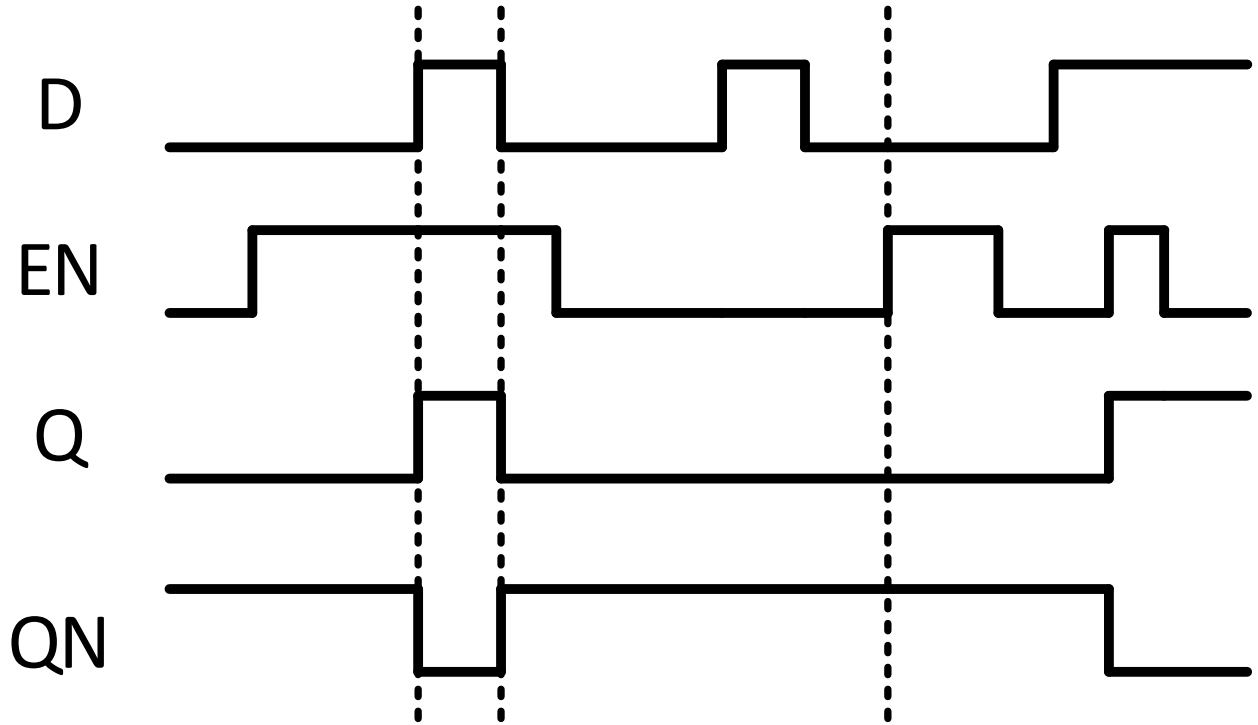
EN	S	R	Q	QN
0	-	-	Q	QN
1	0	0	Q	QN
1	0	1	0	1
1	1	0	1	0
1	1	1	1	1



Latch D com *Enable*



EN	D	Q	QN
0	-	Q	QN
1	0	0	1
1	1	1	0



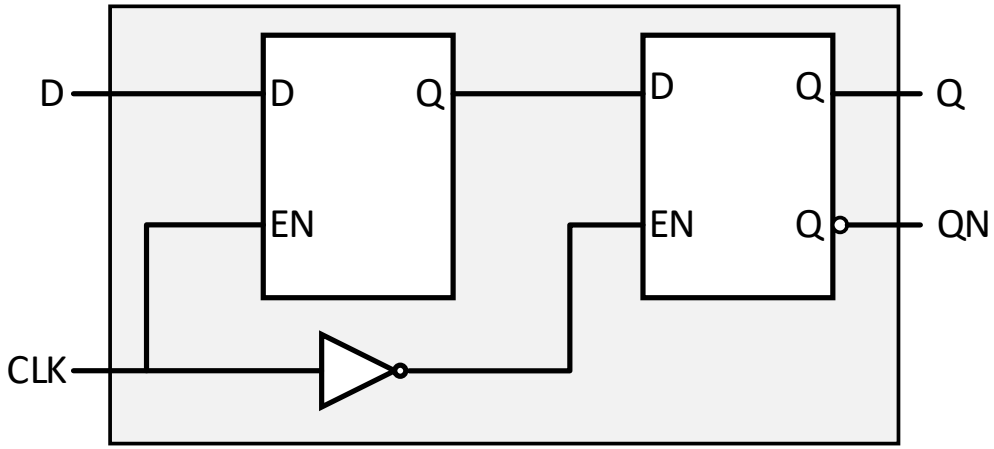
Flip-Flop Master-Slave

Latch – elemento sequencial transparente. Enquanto o *enable* está ativo, as saídas variam com as entradas;

Flip-flop – elemento sequencial não transparente. Eventuais mudanças do estado do elemento ocorrem em flancos de um sinal de sincronização;

Entrada de Relógio – Designação comum da entrada de sincronização.

Edge Triggered D Flip-Flop



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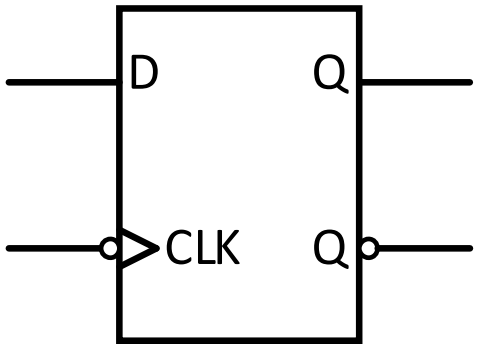
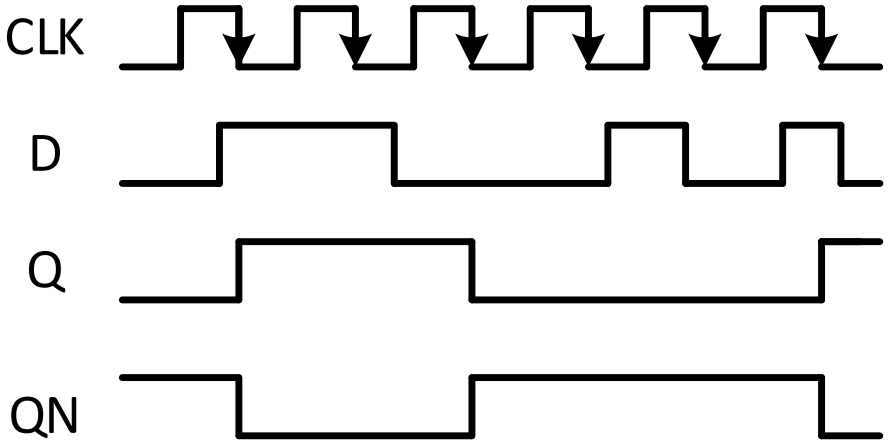


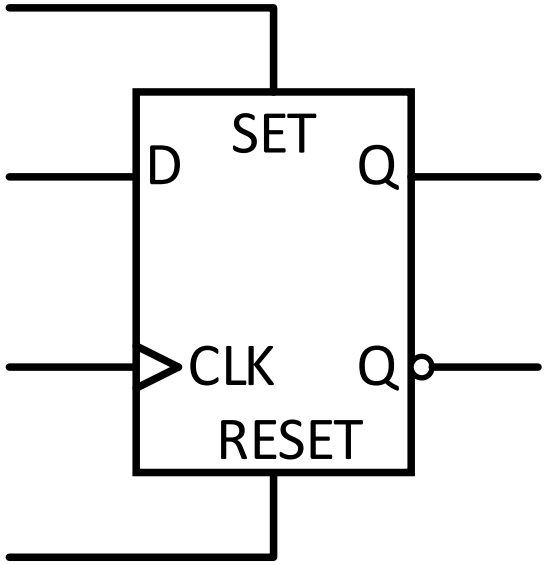
D	CLK	Q	QN
0	↓	0	1
1	↓	1	0
-	0	Q	QN
-	1	Q	QN



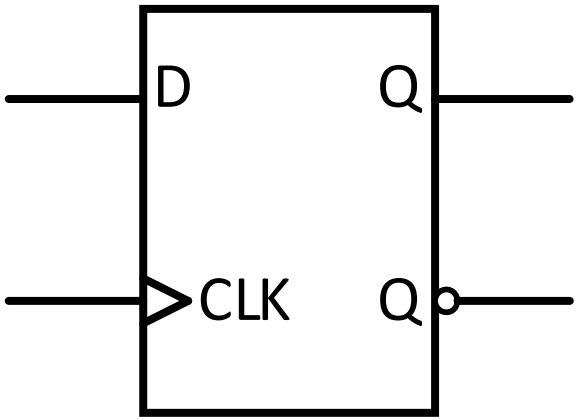
Flip-Flop com Set e Reset

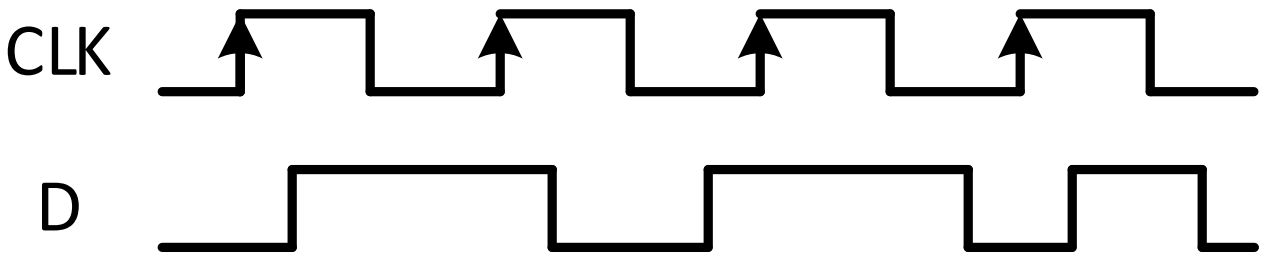
Reset – Coloca a saída Q a '0';
Set – Coloca a saída Q a '1';

Set ou Reset Síncrono – Coloca a saída Q a '1' ou a '0' sincronizado com o sinal de relógio;
Set ou Reset Assíncrono – Coloca a saída Q a '1' ou a '0' de imediato, independentemente do sinal de relógio;



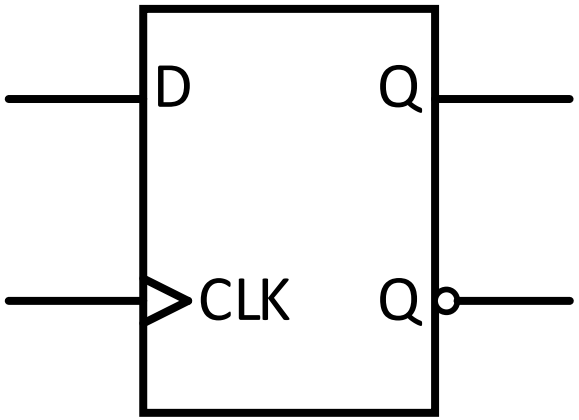
Exercício 1

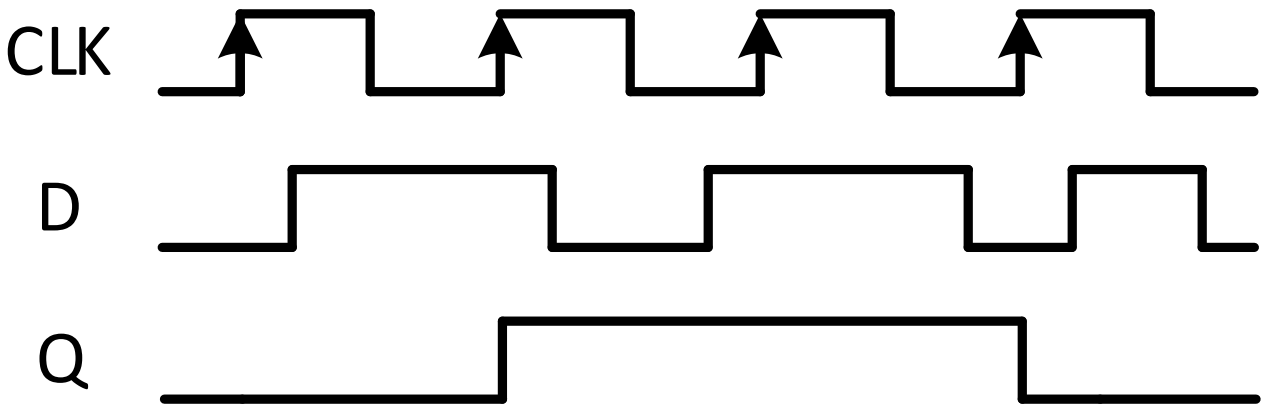




D	CLK	Q	QN
0	↑	0	1
1	↑	1	0

Exercício 1





D	CLK	Q	QN
0	↑	0	1
1	↑	1	0